

MODERN DATA ENGINEERING - FINAL GROUP ASSIGNMENT

Suzlon Energy

End-to-End Data Engineering Implementation on Databricks

Group Members

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1. Objective and Business Context

The objective of this project is to implement the analytical design prepared in Assignment 1 for Suzlon Energy on the Databricks platform. The solution generates realistic synthetic business data, builds a Bronze-Silver-Gold medallion architecture, creates curated KPI tables and dashboard visualizations, and registers an ML-ready feature table for turbine failure-risk analysis.

The project focuses on four business areas: order booking, O&M contract renewal, turbine generation and availability, and service downtime. All data used in this implementation are synthetic and are not intended to represent Suzlon Energy's actual confidential operational or financial data.

2. Synthetic Data Generation

A scenario-driven Python generator was used instead of directly generating CSV rows through an AI tool. The generator creates related customers, sites, turbines, orders, contracts, daily generation records and service events while maintaining business relationships across tables. A fixed random seed supports reproducibility, and the date range and major data volumes are controlled through a configuration file.

2.1 Main business assumptions

- Customer segments include C&I, IPP, PSU and Retail, with different synthetic renewal probabilities.
- Wind-zone quality, turbine capacity, availability, turbine age and seasonality influence daily generation.
- Order value depends on turbine capacity, quantity and customer segment.
- Corrective and emergency service events carry higher synthetic downtime and severity than routine events.
- Generation is created at one row per turbine per day, preserving consistent relationships with turbine, site and date dimensions.

2.2 Generated data volume

Table	Rows
dim_model	15
dim_customer	5,000
dim_site	300
dim_date	365
dim_turbine	1,000
dim_technician	250
fact_order	20,000
fact_contract	6,000
fact_generation	365,000
fact_service	30,000

The largest fact table contains 365,000 daily turbine-generation rows covering 1 January 2025 to 31 December 2025. This provides sufficient scale for trend, ranking, aggregation and window-function analysis.

3. Databricks Medallion Architecture

The generated CSV files were uploaded to a Databricks managed volume and processed through logical Bronze, Silver and Gold layers. Delta tables were used for the engineered layers to provide a reliable tabular storage format and efficient SQL access.

Stage	Implementation	Purpose
Landing	/Volumes/workspace/default/suzlon_landing	Stores generated CSV files before ingestion.
Bronze	bronze_* Delta tables	Preserves the incoming data as received; fields are initially read as strings.
Silver	silver_* Delta tables	Applies explicit schemas, correct data types, cleaning, validation and business-range checks.
Gold	gold_* curated tables	Stores monthly aggregates, rollups and KPI-ready datasets for dashboards and analysis.
Feature Store	feature_turbine_failure	Stores ML-ready turbine-level features registered in Unity Catalog.

3.1 Data-quality and storage decisions

- Bronze preserves raw values and avoids silent type conversion during ingestion.
- Silver converts identifiers to integers, dates to DATE, monetary fields to DECIMAL and operational measures to appropriate numeric types.
- Primary-key and core business-range checks are performed before Gold processing.
- The generation fact is the largest table. Because the project uses a moderate synthetic dataset, the Silver layer is kept physically simple; date-based analysis is handled through the date dimension and smaller Gold monthly aggregates rather than creating many small partitions.

4. Gold Layer and Analytical Processing

The Gold layer contains business-facing aggregates at the correct analytical grain. It supports the dashboard and demonstrates the aggregation and window operations required by the assignment.

Technique / Gold table	Business use
Monthly generation mart	Site-by-month generation and availability.
ROLLUP	Monthly, site/state and overall hierarchical totals.
GROUPING SETS	Selected customer-segment and time-period summaries.
RANK() with PARTITION BY	Ranks turbines within each site by total generation.
3-month moving average	Smooths short-term variation to show the underlying trend.
LAG()	Compares each month with the previous month.

4.1 Dashboard-ready Gold tables

Dedicated Gold tables were created for monthly orders, monthly availability, monthly generation, contract renewal by customer segment, and service downtime by ticket type. This avoids mixing fact tables at incompatible grains and keeps each KPI calculation traceable to a specific curated source.

5. Databricks KPI Dashboard

The dashboard was created using Databricks Dashboards only. Six visuals were selected to show trends, comparisons and rankings rather than only single-number summaries.

Dashboard visual	Business question
Monthly Order Booking Value	How does order-booking value change month by month?
Contract Renewal by Segment	Which customer segments show stronger O&M renewal performance?
Monthly Fleet Availability	Is fleet availability stable, improving or deteriorating over time?
Top 10 Sites by Energy Generation	Which wind sites generate the most energy?
Service Downtime by Ticket Type	Which service categories contribute the most downtime?
Energy Generation Moving Average	What is the underlying generation trend after smoothing short-term variation?

5.1 Dashboard snapshots

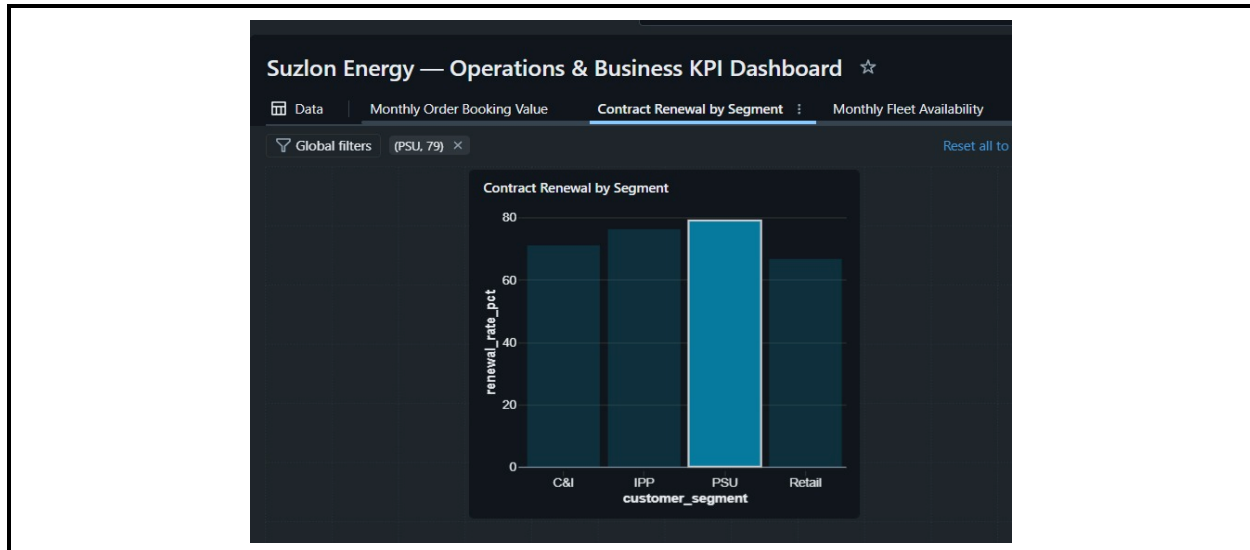
Dashboard 1 - Monthly Order Booking Value

Line chart showing the monthly order-booking trend across 2025.



Dashboard 2 - Contract Renewal by Segment

Bar chart comparing renewal rates across C&I, IPP, PSU and Retail customer segments.



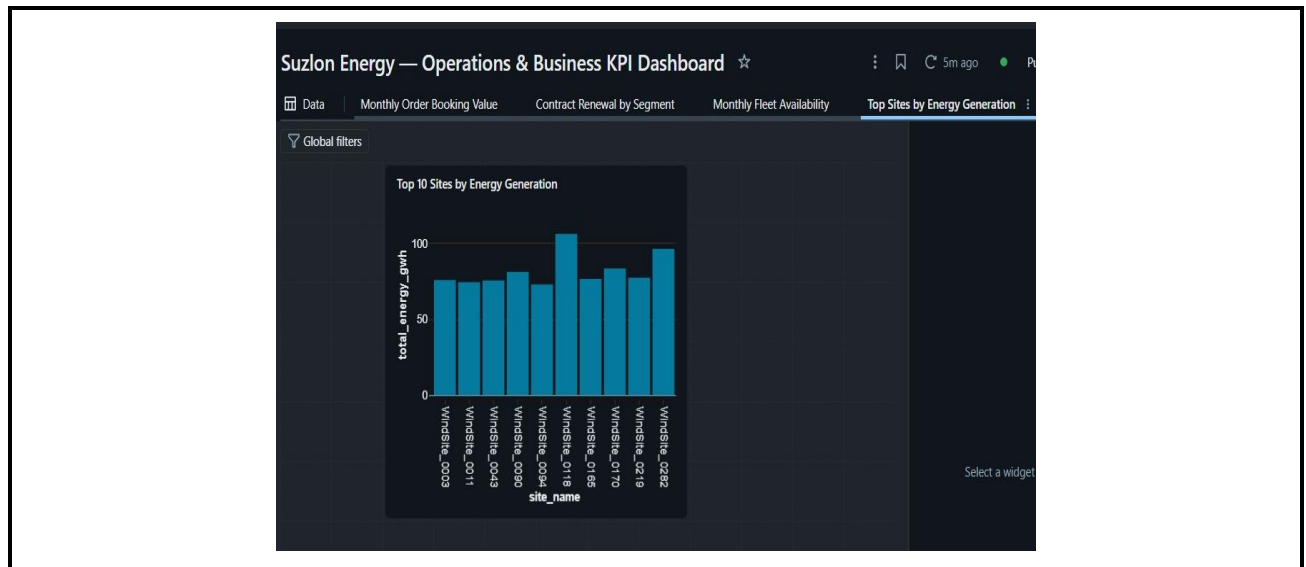
Dashboard 3 - Monthly Fleet Availability

Line chart showing monthly fleet availability during 2025.



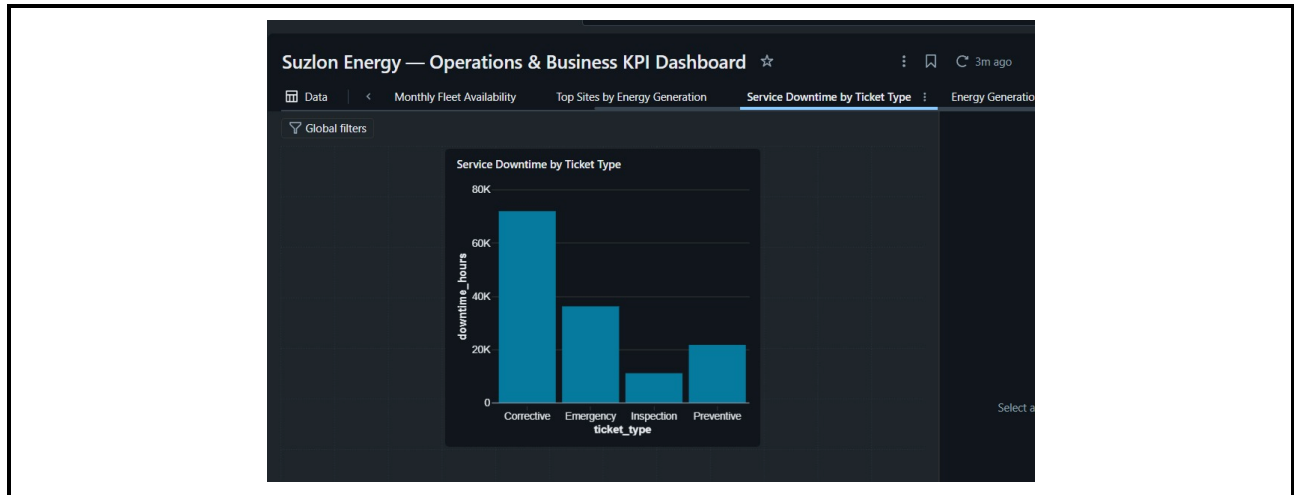
Dashboard 4 - Top 10 Sites by Energy Generation

Bar chart ranking the ten highest-generating wind sites.



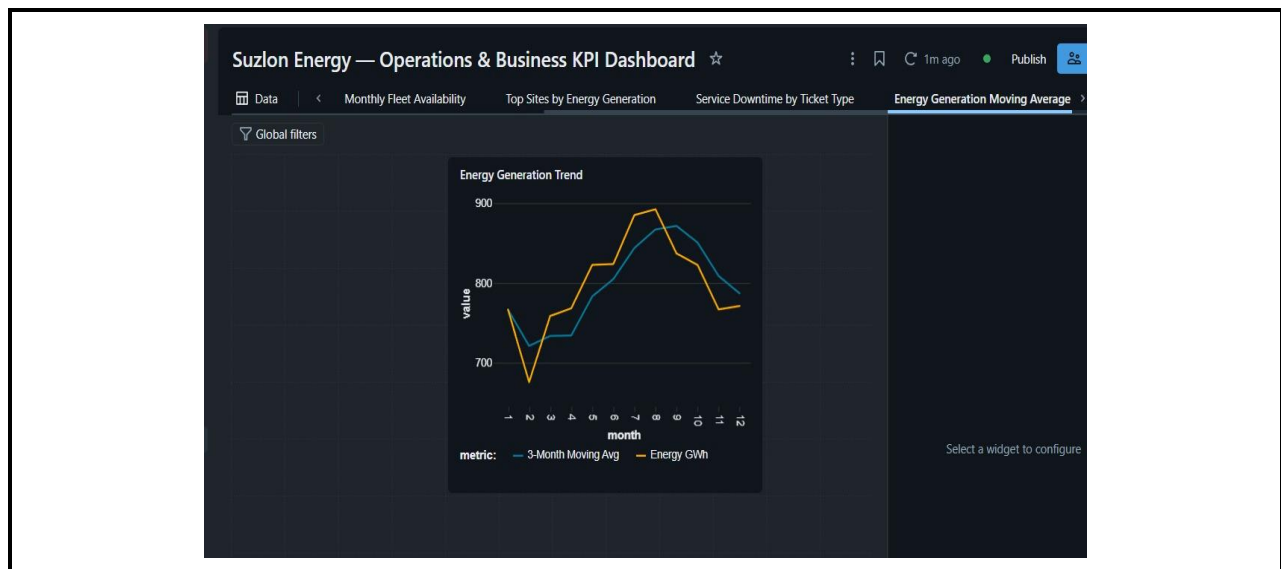
Dashboard 5 - Service Downtime by Ticket Type

Bar chart comparing total downtime for Corrective, Emergency, Inspection and Preventive service events.



Dashboard 6 - Energy Generation Moving Average

Two-line chart comparing monthly energy generation with the 3-month moving average.



6. Feature Store

The ML use case selected from Assignment 1 is turbine/component failure prediction. Because the approved final schema does not contain raw SCADA variables such as vibration or gearbox temperature, the implementation uses operational proxies that are available in the generated data. Features are calculated over the latest 90-day window ending on 31 December 2025.

Feature	Meaning and predictive relevance
avg_availability_pct	Average turbine availability over 90 days; repeated loss of availability may indicate reliability deterioration.
total_downtime_hours_90d	Total downtime over 90 days; higher downtime indicates operational instability.
service_ticket_count_90d	Number of service interventions over 90 days; repeated events may signal elevated failure risk.
avg_energy_kwh_90d	Average daily energy over 90 days; persistent output decline may indicate deteriorating performance.

The feature dataset contains one row per turbine. Quality checks confirmed unique non-null turbine identifiers, valid availability ranges and non-negative operational measures for 1,000 turbines. The table was registered in Unity Catalog as `workspace.default.feature_turbine_failure`, with `turbine_id` as the primary key and `feature_date` as the snapshot date.

7. Key Business Insights

- Monthly order booking remains within a relatively consistent range but shows visible peaks and dips, supporting month-level demand monitoring.
- PSU customers show the strongest synthetic contract-renewal rate, while Retail is comparatively lower, indicating where retention attention may be most useful.
- Fleet availability remains consistently high and stable across the year, with only small month-to-month changes.
- Energy generation differs across sites, supporting direct comparison of top-performing locations.
- Corrective maintenance contributes the largest share of service downtime in the synthetic dataset, followed by emergency events.
- The moving-average view shows a mid-year rise in generation followed by moderation later in the year.

8. Conclusion

The project implements the Assignment 1 analytical design as an end-to-end Databricks workflow. Configurable Python scripts generate the synthetic data; Bronze and Silver layers ingest and clean it; Gold tables support rollups, window functions and six dashboard KPIs; and a registered feature table supports future turbine failure-risk modelling. The implementation therefore connects business intelligence and ML feature engineering within one governed pipeline.